Log Review (Phase 1 Filter Analysis)

Optimize ESC Telemetry Harmonic Notch Before Roll/Pitch PID Tuning

Summary

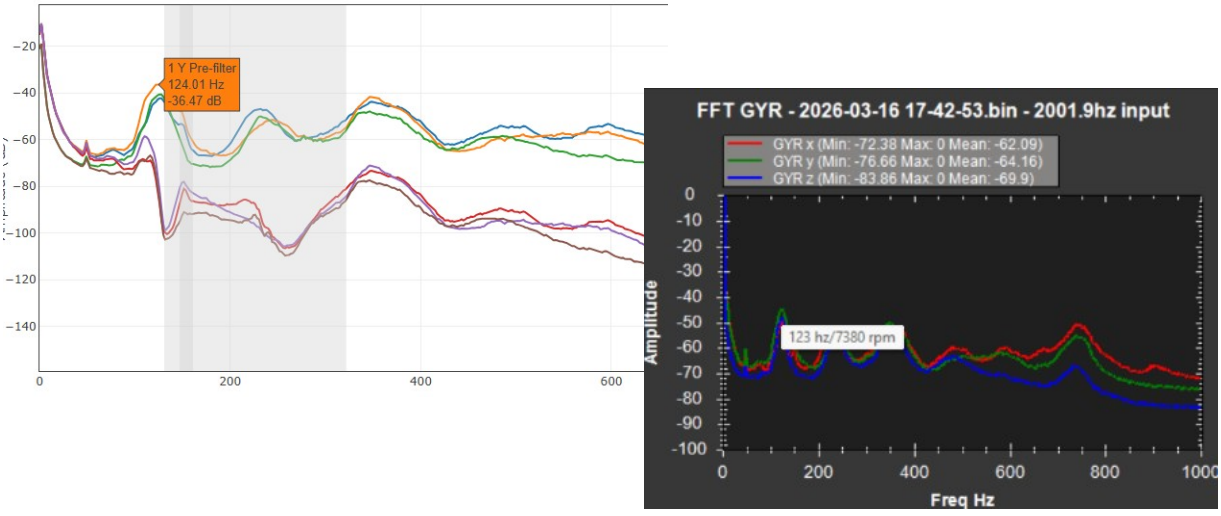
I reviewed the FFT, filter tool, RPM/RCOU, vibration, and roll/pitch plots.

Goal: optimize filter setup before roll/pitch PID work

- **Observed motor band: ~123 Hz**
- **Current notch: 131 Hz / BW 35 / ESC telemetry / multi-source / 3 harmonics**
- **Decision: keep one ESC telemetry notch group, retune it, keep notch #2 disabled**
- **Next-flight changes: `INS_HNTCH_FREQ` 131 → 115, `INS_HNTCH_BW` 35 → 30**
- **Next step: hover + roll/pitch inputs + throttle sweeps, then Phase 2 roll/pitch tuning**

Parameter	Current	Recommended	Why
INS_HNTCH_FREQ	131	115	current base is slightly above the observed hover band
INS_HNTCH_BW	35	30	reduce bandwidth for lower phase lag
INS_HNTC2_ENABLE	0	0	keep second notch off until first notch is validated

Main observation:

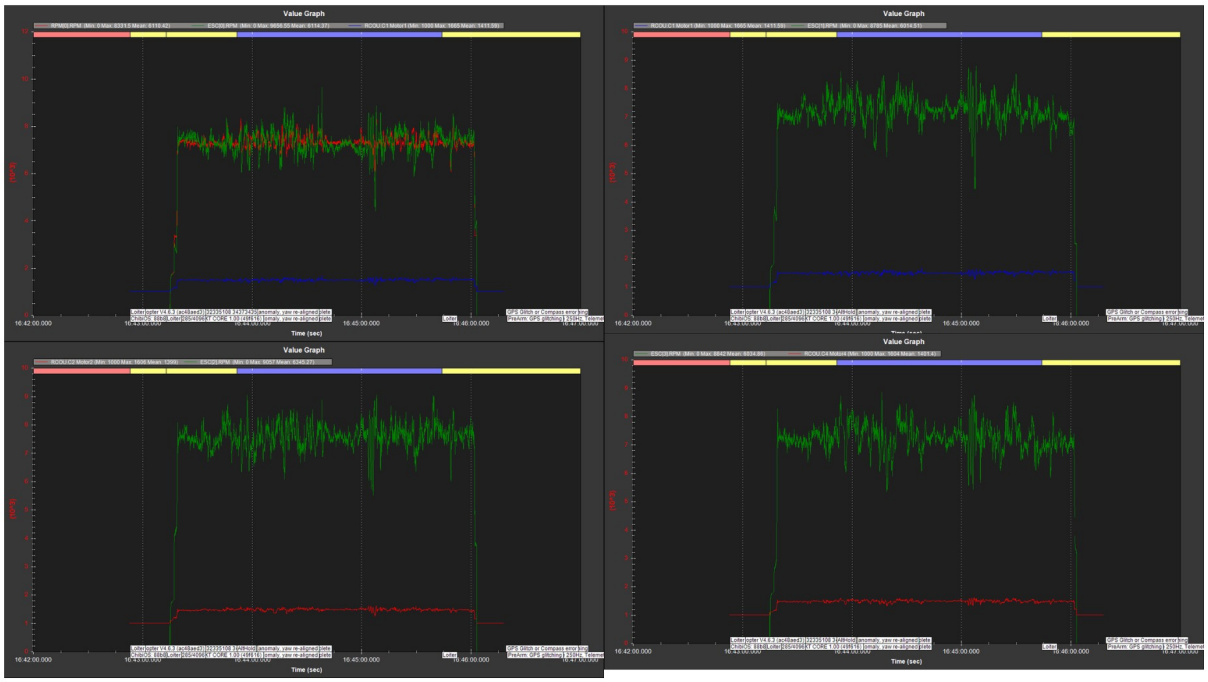


- the main motor noise band is around **~123 Hz**
- current notch base is **131 Hz**
- current bandwidth is **35 Hz**
- second notch group is disabled

Log Analysis : Filtering

First Notch Filter			Second Notch Filter		
INS_HNTCH_ENABLE	1:Enabled		INS_HNTC2_ENABLE	0:Disabled	
INS_HNTCH_MODE	3:ESC Telemetry		INS_HNTC2_MODE	1:Throttle	
INS_HNTCH_FREQ	131	Hz	INS_HNTC2_FREQ	80	Hz
INS_HNTCH_BW	35	Hz	INS_HNTC2_BW	40	Hz
INS_HNTCH_ATT	25	dB	INS_HNTC2_ATT	40	dB
INS_HNTCH_REF	1		INS_HNTC2_REF	0	
INS_HNTCH_FM_RAT	1		INS_HNTC2_FM_RAT	1	

For the next flight, the main improvement is to **retune the first ESC telemetry notch**, not add more PID yet. ArduPilot recommends setting the ESC-telemetry notch base **below hover frequency but above `INS_GYRO_FILTER`**, and narrowing bandwidth when using multi-source tracking to reduce phase lag. ArduPilot also notes that the older double-notch approach is no longer generally recommended.



Review From the plots:

-The filter foundation is good, but the first notch is a bit high for the actual hover motor band. That means for roll and pitch tuning, the next step should be:

1. clean the motor noise band better,
2. re-log hover + roll/pitch inputs,
3. then move to roll/pitch PID changes on the next phase. This matches ArduPilot's tuning flow, where notch setup comes before manual roll/pitch tuning.

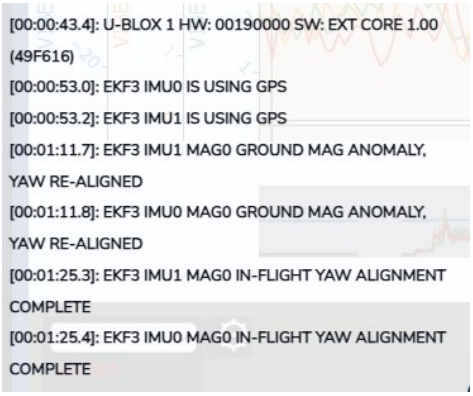
- The oscillation visible in hover looks mainly like a **slow position-control hunt in Loiter**, not a fast inner-loop PID oscillation.

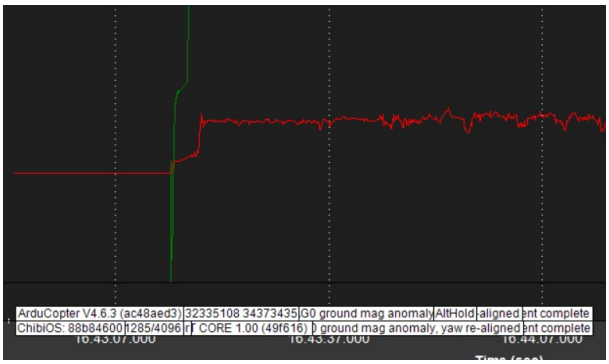
- Since the hover motor band in your data is around the low-120 Hz region, the current `INS_HNTCH_FREQ = 131` is a bit high as a base value. I recommend bringing it slightly lower for the next test.

- The log screenshots also show **ground mag anomaly / yaw realignment** and **GPS glitching** messages. Loiter performance depends heavily on good GPS lock, low magnetic interference, and low vibration. ArduPilot also notes that compass interference can disturb heading and lead to poor position hold behavior.

I do not see this as the main cause of the slow hover oscillation in this test, but it should still be checked before final tuning:

- verify compass calibration,
- verify compass placement / cable routing,
- verify no power wiring or ESC noise is disturbing the compass,
- confirm GPS health before Loiter testing.





Recommendation for Next Flight

<u>Use one optimized ESC telemetry notch group</u>

For this next flight, I do **not** recommend enabling the second harmonic notch group yet.

Reason:

- current setup already has ESC telemetry + multi-source + 3 harmonics
- that is already strong filtering
- adding a second notch group now would add more phase lag before validating the corrected first notch

ArduPilot's guidance is to optimize the main dynamic notch first, then only add more filtering if the next log still shows remaining motor-band noise.

Parameter Changes for Next Flight

- **INS_GYRO_FILTER = 80**
- **INS_ACCEL_FILTER = 20**
- **SCHED_LOOP_RATE = 400**
- **INS_HNTCH_ENABLE = 1**

- INS_HNTCH_MODE = 3
- INS_HNTCH_FREQ = 115
- INS_HNTCH_BW = 30
- INS_HNTCH_ATT = 25
- INS_HNTCH_REF = 1
- INS_HNTCH_FM_RAT = 1
- INS_HNTCH_HMNCS = 3
- INS_HNTCH_OPTS = 6
- INS_HNTC2_ENABLE = 0

Main changes only

- INS_HNTCH_FREQ : 131 -> 115
- INS_HNTCH_BW : 35 -> 30
- INS_HNTC2_ENABLE = 0 (keep disabled)

What stays unchanged for this flight

- ATC_RAT_RLL_P
- ATC_RAT_RLL_I
- ATC_RAT_RLL_D
- ATC_RAT_PIT_P
- ATC_RAT_PIT_I
- ATC_RAT_PIT_D

Next Flight Instructions

Use this as a **filter validation + roll/pitch foundation flight**:

1. Take off in **Stabilize** or **AltHold**
2. Hold a steady hover for **30 seconds**
3. Do **3 to 4 short roll inputs**
4. Do **3 to 4 short pitch inputs**
5. Do **2 to 3 small throttle sweeps**
6. Land and save full log

ArduPilot recommends at least a short stable hover log for notch evaluation, and reducing gyro noise first improves the quality of the tune.

What we want from the next log

- cleaner post-filter gyro around **110–140 Hz**
- better roll and pitch rate cleanliness
- no new fast oscillation
- then we move to **Phase 2: roll/pitch PID tuning**